ECE4010J Probabilistic Methods in Engineering (Summer 2026)

Assignment 1

Date Due: See canvas

This assignment has a total of **(140 points)**.

Note: Unless specified otherwise, you must show the details of your work via logical reasoning for each exercise. Simply writing a final result (whether correct or not) will receive **0 point**.

Exercise 1.1 (20 pts) In a bridge game, what is the probability that the bridge hands of East and West together contain exactly k kings, where $k = 0, 1, 2, 3, 4$?

Exercise 1.2 (10 pts) Let A_1 and A_2 be mutually exclusive events such that $\mathbb{P}[A_1]\mathbb{P}[A_2] > 0$. Show that these events are not independent.

Exercise 1.3 (10 pts) Let A_1 and A_2 be independent events such that $\mathbb{P}[A_1]\mathbb{P}[A_2] > 0$. Show that these events are not mutually exclusive.

Exercise 1.4 (10 pts) As society becomes dependent on computers, data must be communicated via public communication networks such as satellites, microwave systems, and telephones. When a message is received, it must be authenticated. This is done by using a secret enciphering key. Even though the key is secret, there is always the possibility that it will fall into the wrong hands, thus allowing an unauthentic message to appear to be authentic. Assume that 95% of all mesages received are authentic. Furthermore, assume that only 0.1% of all unauthentic messages are sent using the correct key and that all authentic messages are sent using the correct key. Find the probability that a message is authentic given that the correct key is used.

Exercise 1.5 (90 pts) Verify the binomial coefficient formulas using algebraic methods (no stories). m, n, k are integers.

$$\binom{n}{k} = \binom{n}{n-k}, \qquad n \geq 0$$

$$\binom{r}{k} = \frac{r}{k}\binom{r-1}{k-1}, \qquad k \neq 0$$

$$\binom{r}{k} = \binom{r-1}{k} + \binom{r-1}{k-1}$$

$$\binom{r}{k} = (-1)^k \binom{k-r-1}{k}$$

$$\binom{r}{m}\binom{m}{k} = \binom{r}{k}\binom{r-k}{m-k}$$

$$\sum_{k \leq n} \binom{r+k}{k} = \binom{r+n+1}{n}$$

$$\sum_{k=0}^{n} \binom{k}{m} = \binom{n+1}{m+1}, \qquad m, n \geq 0$$

$$\frac{1}{(1-z)^{n+1}} = \sum_{k \geq 0} \binom{n+k}{n} z^k, \qquad n \geq 0$$

$$\frac{z^n}{(1-z)^{n+1}} = \sum_{k \geq 0} \binom{k}{n} z^k, \qquad n \geq 0$$